Ablation of Hybrid Correction Strategies for Minority-Class Misclassification

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Abstract*—Imbalanced datasets cause classifiers to miss minority-class examples more often than they should, and this becomes a real problem when those misses carry much higher costs than false alarms. Three main correction approaches exist: resampling the training data, adding class-level penalties to the loss function, and using ensemble classifiers that can handle skewed distributions better by design. In practice these methods are often used together, yet it is still not clear whether combining them adds value or whether they work against one another, since this has not been examined in a controlled way. We ran an ablation study using the Pima Indians Diabetes Database (34.9% positive class) to test all pairwise and full combinations of SMOTETomek resampling (Strategy A), class weighting (Strategy B), and XGBoost (Strategy C) across five classifiers and six metrics. The clearest result is that applying class weights to the original imbalanced data, without any resampling, gives the best XGBoost performance: macro-F1 of 0.7294 and ROC-AUC of 0.8306. Adding resampling on top of this makes things worse, not better. When both the training distribution and the loss function push toward the minority class at the same time, precision drops without any gain in recall, and macro-F1 ends up lower.

Index Terms*—class imbalance, minority-class misclassification, SMOTE, Tomek Links, class weights, XGBoost, ablation study, imbalanced learning*

I. INTRODUCTION

Class imbalance is one of the most common structural problems in real-world machine learning. When training labels are heavily skewed, a standard classifier will tend to favor the majority class, since minimizing overall loss rewards that behavior, and the rare class gets mostly ignored [1]. This matters especially in fields like medical diagnosis, fraud detection, and equipment fault monitoring, where a missed positive case carries far higher consequences than a false alarm.

The problem goes beyond just having fewer positive examples. It is also geometric. The decision boundary a classifier learns from skewed data gets pushed toward the majority class, which cuts into its ability to detect the minority class even when those examples have clear patterns [2]. Fixing this requires changing either what data the classifier trains on, how it penalizes mistakes, or both.

Three types of correction have been studied separately in the literature. Data-level methods such as SMOTE [1] and Tomek Links [3] work on the training set before learning starts, either by generating synthetic minority examples, removing majority ones, or cleaning up the class boundary. Algorithm-level methods assign higher misclassification costs to the minority class directly inside the loss function. Ensemble classifiers, particularly gradient-boosted trees like XGBoost [4], build corrections across many iterations and can tolerate some degree of imbalance without needing an explicit fix.

Each of these approaches has been studied on its own, but what happens when they are combined has received much less attention. Practitioners often stack multiple strategies together, yet it is unclear which part of the combination is doing the actual work, or whether some strategies actually conflict with each other under certain conditions. Without a direct controlled comparison, picking the right strategy for a given dataset is largely guesswork, particularly for tabular data with moderate imbalance.

This paper directly tests that question. We define three strategies and run all four possible pairings (A+B, B+C, A+C, and A+B+C) across five classifiers on six evaluation metrics, using a held-out test set. The Pima Indians Diabetes Database provides a clean, well-understood benchmark for moderate binary imbalance. While the results come from a diabetes dataset, the findings are meant to speak to the broader imbalanced classification problem, not to diabetes prediction specifically.

This work makes three contributions. First, it provides a controlled ablation design that measures what each correction strategy actually adds, both alone and in combination. Second, it shows empirically that algorithm-level class weighting without resampling outperforms all combined configurations, including the full three-way hybrid. Third, it explains the mechanism behind this result: a precision-recall imbalance that develops when data-level and loss-level corrections are both applied at the same time.

The rest of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and preprocessing steps. Section IV explains the proposed methodology. Section V reports the results. Section VI discusses what they mean. Section VII presents the conclusion.

II. RELATED WORK

A. Algorithm-Level Correction Without Resampling

Jabbar et al. [5] looked at a heavily imbalanced Iraqi diabetes dataset where diabetic cases outnumbered pre-diabetic ones at 690:40. Rather than touching the training data, the authors embedded higher error penalties directly into cost-sensitive XGBoost and CART models, giving pre-diabetic false negatives double the weight of other mistakes. This alone brought balanced accuracy to 93.84%, showing that algorithm-level correction can work well without generating synthetic data. The setup is close enough to our Test 2 configuration that their result serves as an external reference point for algorithm-only correction, though their imbalance ratio is far more extreme than ours.

B. Resampling with Boundary Cleaning

Mia et al. [6] handled class imbalance in blood cell image classification by first running SMOTE and then applying Tomek Links, before passing the data to a Dual Path Sliding Window Attention network. The reasoning is straightforward: SMOTE sometimes places synthetic samples right at the class boundary, which adds noise rather than useful signal; Tomek Links cleans this up by finding and dropping inter-class border pairs. The full pipeline reached 99.02% accuracy, and the reported gain over SMOTE-only conditions supports using SMOTETomek as Strategy A in this work. A recent 2026 review by Nikpour et al. [10] surveys data-level methods for imbalanced classification and confirms that boundary-aware resampling such as SMOTETomek remains a strong and widely used choice, which further motivates its use here.

C. Multi-Strategy Hybrid for High-Stakes Detection

Rahman et al. [7] tackled TB detection from chest X-rays by stacking four corrections: SMOTETomek resampling, Modified Focal Loss, class weighting, and an EfficientNetB0/DenseNet121 ensemble fused with Random Forest. The system reached 99.7% accuracy and 98% recall on TB-positive cases. The setting is very different from ours (image data, deep learning, four-part combination), but it is still relevant because it shows how high full strategy combination can push performance, while never breaking down how much each individual component contributed. That decomposition is exactly what this ablation study attempts to provide.

D. SMOTE with XGBoost on Tabular Data

Siagian et al. [8] evaluated SMOTE-balanced XGBoost on two tabular datasets, including the Pima Indians Diabetes Database used in this study, reporting F1-score of 0.77 and ROC-AUC of 0.83 for the diabetes classification task. This work serves as our primary baseline. We extend it in two material respects: adding Tomek Links cleaning to form the complete SMOTETomek pipeline and, more importantly, situating SMOTE+XGBoost as one cell in a controlled ablation matrix rather than a standalone contribution.

E. Research Gaps

The work reviewed above shows that resampling, algorithm-level weighting, and ensemble classifiers can each handle minority-class misclassification reasonably well when tested in isolation. What is missing from all of them is a direct comparison of the combinations, all run on the same data with the same evaluation setup. Without that, there is no way to tell which pairings actually help, which are redundant, and which end up working against each other. This study is set up to answer exactly that. A recent 2026 study by Sakho et al. [11] adds weight to this concern. Working on tabular data, they show that for many datasets applying no resampling at all is competitive with SMOTE and its variants. This is the same question our ablation targets, that is, whether stacked corrections truly help or simply add cost.

III. BENCHMARK DATASET AND PREPROCESSING

A. Dataset Selection and Properties

The Pima Indians Diabetes Database [9] is a widely used benchmark for binary classification under class imbalance. It contains 768 patient records, each with eight numeric features: number of pregnancies, plasma glucose level, diastolic blood pressure, triceps skin fold thickness, serum insulin, BMI, a diabetes pedigree score, and age. The outcome label is binary (1 for diabetic, 0 for non-diabetic), with 500 negative cases and 268 positive ones, giving a positive-class ratio of 34.9%. This falls in the moderate-imbalance range that is common in tabular clinical data. The imbalance is real enough that different correction strategies produce noticeably different results across classifiers, but not so severe that one approach dominates by default. This is the kind of setting where an ablation study is most useful.

Fig. 1: Class distribution of the benchmark dataset: 500 negative instances and 268 positive instances (minority ratio 34.9%).

B. Missing Value Treatment

Five features (glucose, blood pressure, skin thickness, insulin, and BMI) contain physiologically implausible zero values. These are treated as missing and imputed using column-wise median values, a choice that is robust to the skewed distributions present in several of these features. Following imputation, no missing values remain in the dataset.

C. Feature Analysis

Figure 2 illustrates per-feature distributions stratified by class. Plasma glucose and BMI exhibit the most pronounced inter-class separation, consistent with their established clinical relevance to diabetes. Figure 3 presents the lower-triangular correlation matrix; glucose carries the highest linear correlation with the outcome ($r = 0.47$), followed by BMI ($r = 0.29$) and age ($r = 0.24$). The moderate inter-feature correlations indicate that multicollinearity is unlikely to compromise classifier training.

D. Normalization and Partitioning

All features are scaled to [0, 1] using Min-Max normalization before any resampling is applied. This step is needed because SMOTE uses Euclidean distance to find nearest neighbors during interpolation. Features with large numeric ranges would otherwise dominate the distance calculation and pull synthetic samples into unrepresentative parts of the feature space.

The data is split 80/20 into training and test sets using stratified sampling with a fixed random seed, giving 614 training instances (400 negative, 214 positive) and 154 test instances (100 negative, 54 positive). The test set is kept entirely separate until final evaluation; it plays no part in resampling, cross-validation, or classifier selection at any point during the experiments.

IV. PROPOSED METHODOLOGY

This section describes how the three correction strategies are implemented and how they are combined into the four ablation configurations evaluated in this study. The same preprocessing, classifier pool, and evaluation protocol are applied consistently across every configuration, so that any difference in outcome can be attributed to the strategy combination itself rather than to inconsistencies elsewhere in the pipeline. Fig. 4 shows the overall architecture of the pipeline, from the raw dataset to the final ablation comparison.

A. Strategy A: SMOTETomek Resampling

SMOTE [1] generates synthetic minority-class instances through linear interpolation. For each minority sample xi and a randomly selected neighbor xnn from its k nearest minority peers, a synthetic point is placed at:

$$x_{new} = x_i + \lambda (x_{nn} - x_i), \quad \lambda \sim U(0, 1). \qquad (1)$$

A known limitation of (1) is that interpolated samples may fall within majority-class territory, creating overlapping regions that degrade boundary sharpness. Tomek Links [3] addresses this by identifying inter-class nearest-neighbor pairs (instances from opposite classes that are each other's closest neighbors) and removing the synthetic member of each such pair. The combined SMOTETomek pipeline therefore simultaneously enriches the minority-class region and prunes its noisiest boundary samples. Applied to the training partition, SMOTETomek produces a rebalanced set of 385 instances per class (down from the oversampled maximum, reflecting the boundary-cleaning removal of 15 majority samples).

B. Strategy B: Class Weighting

Class weighting introduces asymmetric loss penalties at the optimization level. For a training instance from class c, its contribution to the loss is scaled by:

$$w_c = N / (K \cdot N_c), \qquad (2)$$

where N is the total training sample count, K is the number of classes, and Nc is the count for class c. For the original training partition ($N = 614$, $N_0 = 400$, $N_1 = 214$, $K = 2$), this yields $w_0 = 0.7675$ and $w_1 = 1.4346$. For XGBoost, the equivalent parameter is scale_pos_weight = $N_0/N_1 = 1.8692$.

One design decision is worth noting for Tests 1 and 4, where both resampling and class weighting are active: the class weights are calculated from the original imbalanced distribution and left unchanged when applied to the resampled data. This is intentional. Weights calibrated for the original 1.87:1 ratio will over-correct when the training distribution is already close to balanced. The size of that effect can be measured directly by comparing Test 1 (A+B) against Test 3 (A+C), which uses resampling alone.

C. Strategy C: XGBoost Ensemble

XGBoost [4] builds decision tree ensembles using gradient boosting with second-order loss approximation. Regularization and column subsampling keep it from overfitting, and its residual-correction mechanism across iterations gives it some natural ability to pick up minority-class patterns without an explicit imbalance fix. Hyperparameters are held fixed at n_estimators=300, max_depth=4, learning_rate=0.05, subsample=0.8, and colsample_bytree=0.8 across all four tests.

D. Ablation Test Matrix

Table I lists the four test configurations. Each row activates a different combination of strategies, while the classifier pool, preprocessing steps, and evaluation setup stay the same across all four tests. Any performance difference between tests should therefore come from the strategy combination, not from anything else in the pipeline.

Each test configuration is also run on four other classifiers: Decision Tree (DT), Support Vector Machine (SVM), Gaussian Naive Bayes (GNB), and Logistic Regression (LR). This checks whether the findings carry across different model types. Before evaluating on the held-out set, ten-fold stratified cross-validation is run on the training partition.

TABLE I: ABLATION STUDY TEST CONFIGURATIONS

Test	SMOTE+Tomek	Class Weights	XGBoost	Active Correction
1 (A+B)	✓	✓		Resample + Weight
2 (B+C)		✓	✓	Weight only
3 (A+C)	✓		✓	Resample only
4 (A+B+C)	✓	✓	✓	All three

E. Evaluation Metrics

Each model is evaluated on the held-out test set using six metrics: accuracy, per-class precision and recall, per-class F1-score, macro-averaged F1 (macro-F1), and ROC-AUC. Macro-F1 is the primary metric for comparing tests. Unlike accuracy, it averages F1 equally across both classes, so a strategy that improves the majority class at the minority's expense will score lower, and that is exactly the failure mode imbalance correction is supposed to avoid.

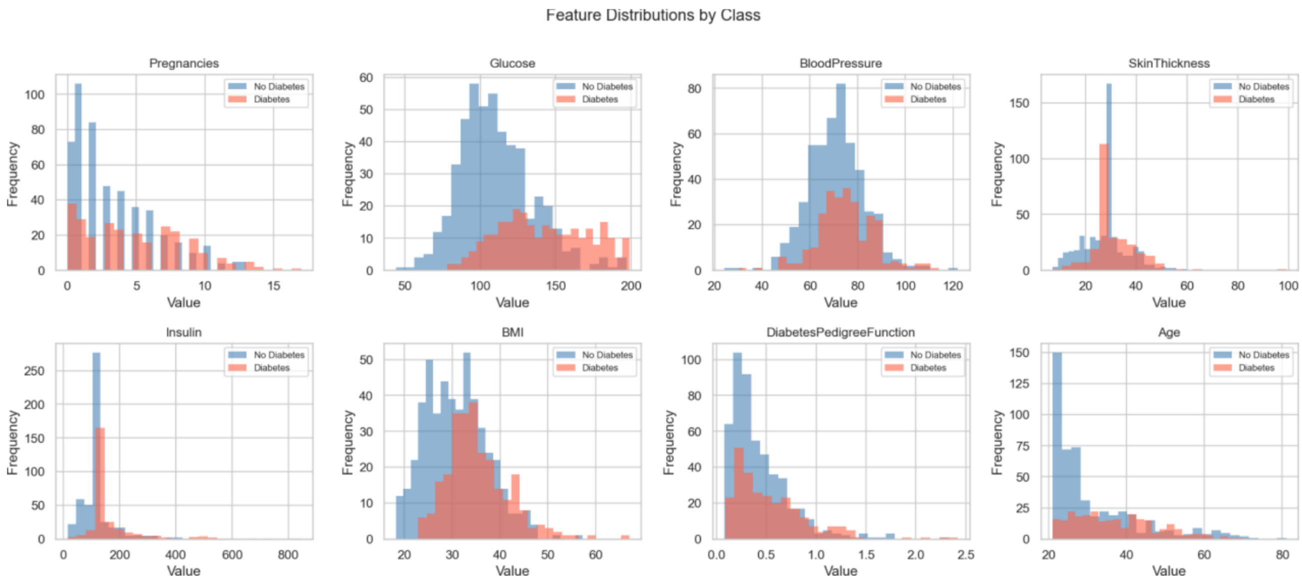

Fig. 2: Per-feature distributions stratified by class label. Glucose and BMI exhibit the strongest class-conditional separation.

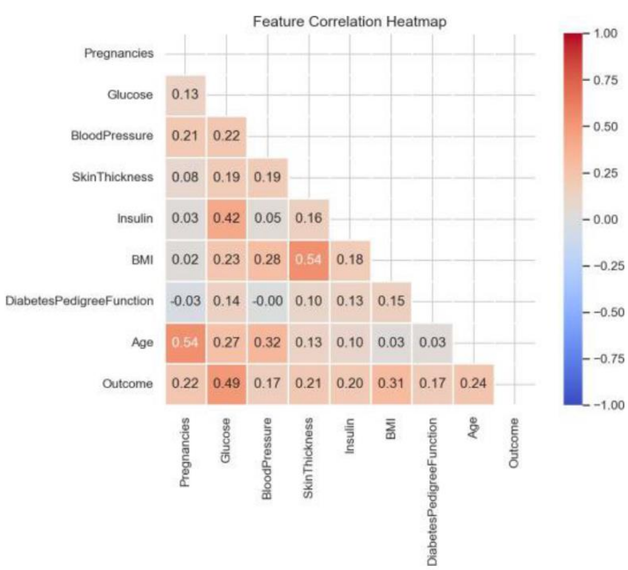

Fig. 3: Lower-triangular Pearson correlation matrix. Glucose presents the strongest linear association with the outcome (r = 0.47).

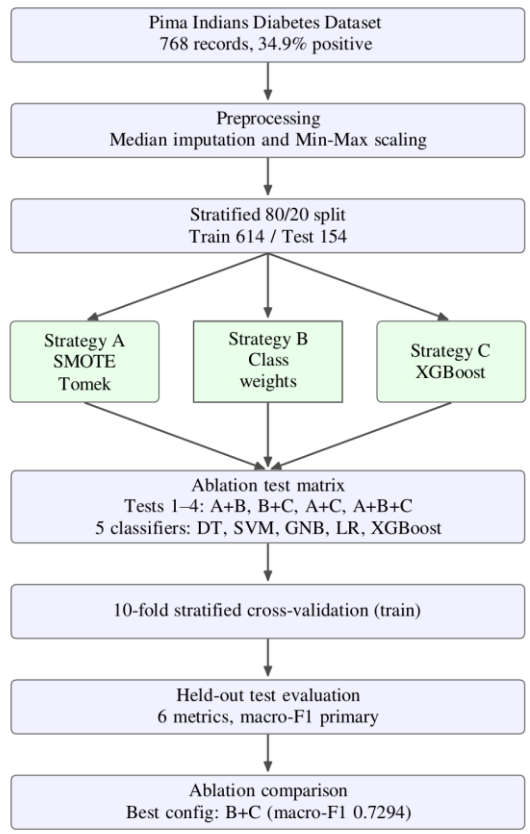

Fig. 4: Overall architecture of the ablation pipeline. The three correction strategies (A, B, C) are combined into the four test configurations, run across five classifiers, and compared on the held-out test set.

V. RESULTS

A. Test 1: SMOTETomek with Class Weights (A+B)

Putting original-distribution class weights on top of already-balanced SMOTETomek data raises positive-class recall across all classifiers. LR reaches R1 = 0.8889, but precision takes a clear hit. XGBoost lands at macro-F1 of 0.6937 and ROC-AUC of 0.8137. This matches what the over-correction hypothesis predicts: when both the training data and the loss function are simultaneously pushing toward the minority class, the decision boundary shifts too far, giving higher recall at the cost of precision and ultimately a lower macro-F1.

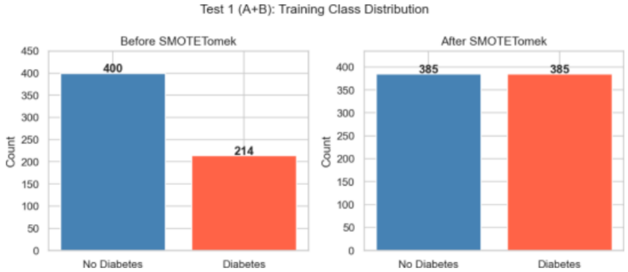

Fig. 5: Training class distribution before and after SMOTETomek. Minority instances increase from 214 to 385; 15 borderline majority instances are concurrently removed by Tomek Links.

TABLE II: TEST 1 (A+B): SMOTETOMEK + CLASS WEIGHTS

Model	Acc.	P1	R1	F11	MF1
DT	0.7078	0.6000	0.5000	0.5455	0.6651
SVM	0.7013	0.5490	0.8704	0.6731	0.6988
GNB	0.6688	0.5200	0.7593	0.6176	0.6626
LR	0.6948	0.5400	0.8889	0.6716	0.6932
XGBoost	**0.7078**	**0.5672**	**0.7037**	**0.6281**	**0.6937**

B. Test 2: Class Weights with XGBoost (B+C)

Using class weights on the original imbalanced data, with no other changes, gives the best XGBoost numbers across all four tests: macro-F1 of 0.7294 and ROC-AUC of 0.8306. What stands out here is how this result is achieved: XGBoost gets higher precision (P1 = 0.6230 vs. 0.5672 in Test 1) while recall stays exactly the same (R1 = 0.7037 in both cases). The precision gain comes from not over-correcting the training distribution. SVM also holds up well at macro-F1 of 0.7253, which suggests this is not a result specific to XGBoost.

TABLE III: TEST 2 (B+C): CLASS WEIGHTS WITHOUT RESAMPLING

Model	Acc.	P1	R1	F11	MF1
DT	0.7078	0.6176	0.4444	0.5172	0.6534
SVM	0.7338	0.5938	0.7963	0.6804	0.7253
GNB	0.6818	0.5373	0.6481	0.5875	0.6645
LR	0.7143	0.5806	0.6667	0.6207	0.6958
XGBoost	**0.7468**	**0.6230**	**0.7037**	**0.6609**	**0.7294**

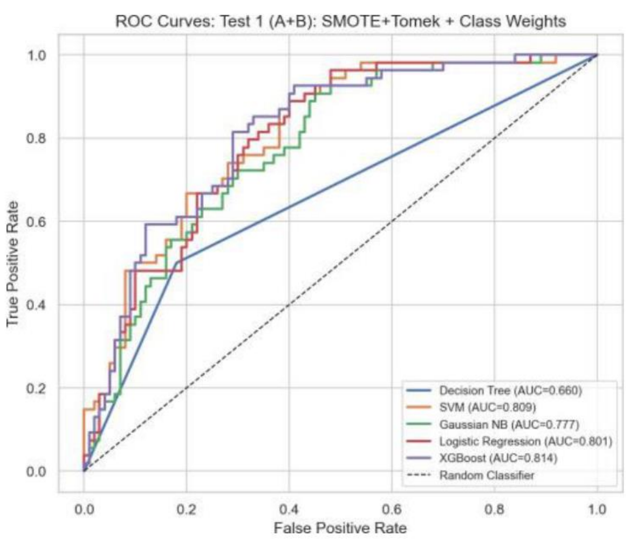

Fig. 6: Test 2 (B+C) diagnostics for the five classifiers. Top: ROC curves, where XGBoost reaches the highest AUC of 0.831.

C. Test 3: SMOTETomek with XGBoost (A+C)

Resampling without class weighting falls between Tests 2 and 1 in performance. XGBoost reaches macro-F1 of 0.7115 and ROC-AUC of 0.8104. One notable result here is the Decision Tree, which achieves its best macro-F1 across all four configurations (0.7038) under Test 3. This likely reflects that Tomek Links boundary cleaning is especially useful for tree induction, which relies directly on finding clean decision thresholds. Compared to the baseline in [8] that uses plain SMOTE without boundary cleaning, Test 3 shows a measurable improvement, so Tomek Links is contributing meaningfully here.

TABLE IV: TEST 3 (A+C): SMOTETOMEK WITHOUT CLASS WEIGHTS

Model	Acc.	P1	R1	F11	MF1
DT	0.7338	0.6279	0.5926	0.6098	0.7038
SVM	0.7403	0.6136	0.7222	0.6635	0.7252
GNB	0.7013	0.5606	0.6852	0.6167	0.6860
LR	0.7078	0.5714	0.6667	0.6154	0.6899
XGBoost	**0.7273**	**0.5938**	**0.7037**	**0.6441**	**0.7115**

D. Test 4: Full Hybrid (A+B+C)

Tests 1 and 4 give identical numbers for every classifier. This is a structural result: both use the same resampled data and the same original-distribution class weights, and XGBoost is already included as a candidate in every test. Adding it explicitly as a strategy in Test 4 changes nothing. The key takeaway from this equivalence is that the over-correction seen in Test 1 is not something XGBoost can compensate for. It is baked into the data and weight configuration shared by both tests.

TABLE V: TEST 4 (A+B+C): FULL HYBRID CONFIGURATION

Model	Acc.	P1	R1	F11	MF1
DT	0.7078	0.6000	0.5000	0.5455	0.6651
SVM	0.7013	0.5490	0.8704	0.6731	0.6988
GNB	0.6688	0.5200	0.7593	0.6176	0.6626
LR	0.6948	0.5400	0.8889	0.6716	0.6932
XGBoost	**0.7078**	**0.5672**	**0.7037**	**0.6281**	**0.6937**

TABLE VI: XGBOOST PERFORMANCE ACROSS ALL FOUR TESTS (CORE ABLATION)

Test	Acc.	MF1	AUC	P1	R1	F1
1 (A+B)	0.7078	0.6937	0.8137	0.5672	0.7037	0.62
2 (B+C)	0.7468	0.7294	0.8306	0.6230	0.7037	0.66
3 (A+C)	0.7273	0.7115	0.8104	0.5938	0.7037	0.64
4 (A+B+C)	0.7078	0.6937	0.8137	0.5672	0.7037	0.62

E. Cross-Test Ablation Comparison

XGBoost's positive-class recall is fixed at 0.7037 across all four tests. It correctly identifies 38 of the 54 positive test instances in every configuration. What separates the tests is precision: Test 2 achieves the highest P1 = 0.6230, which is 5.5 percentage points above Tests 1/4 and 2.9 above Test 3. The macro-F1 ordering B+C > A+C > A+B = A+B+C tracks this precision ordering directly, confirming that the over-correction effect works through precision, not through a shift in the decision threshold.

VI. DISCUSSION

A. The Over-Correction Phenomenon

Class weighting on the original imbalanced data (Test 2: B+C) beats every other configuration, including the full hybrid. The cause is a concrete mismatch: SMOTETomek moves the training ratio to about 1:1, but the class weights were still set for the original 1.87:1 ratio, so they push too hard toward the minority class on already-balanced data. Recall stays fixed at 0.7037, while precision drops from 0.6230 (Test 2) to 0.5672 (Tests 1 and 4) for XGBoost and from 0.5938 to 0.5490 for SVM. Because macro-F1 tracks this precision drop, it falls from 0.7294 to 0.6937. The same drop appears across all five classifiers, so it comes from how the two corrections interact, not from any single algorithm.

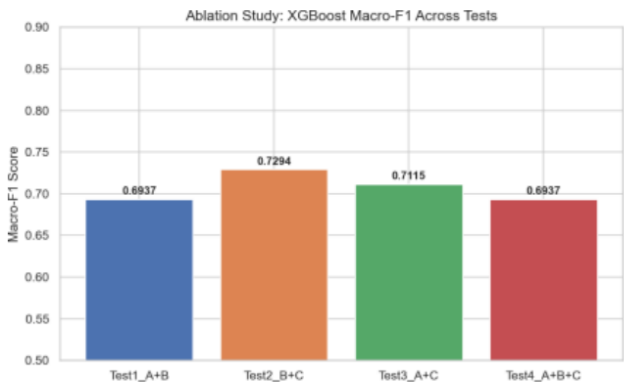

Fig. 7: XGBoost macro-F1 across all four ablation tests. Test 2 (B+C) achieves the highest score (0.7294); Tests 1 and 4 are equivalent.

B. Resampling as an Independent Correction

SMOTETomek alone (Test 3) reaches macro-F1 of 0.7115 for XGBoost, above Tests 1 and 4 (0.6937) and above the plain SMOTE baseline in [8]. Since Test 3 uses no class weights yet still beats the weighted-and-resampled Tests 1 and 4, the compounded weights in those tests are the factor that lowers performance, not a missing correction.

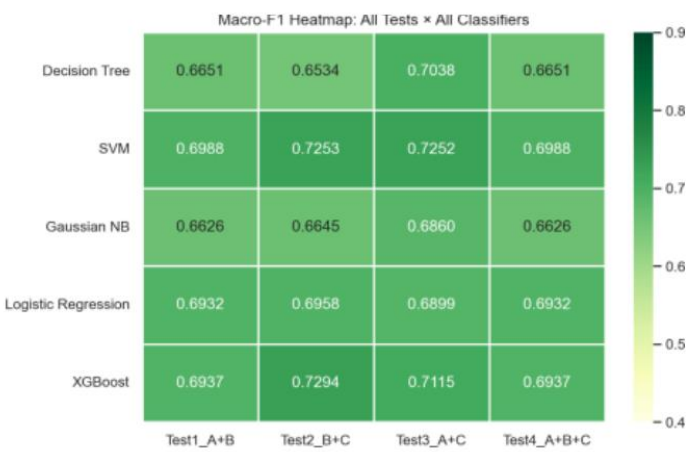

Fig. 8: Macro-F1 heatmap (classifiers × tests). Test 2 (B+C) dominates for the three strongest classifiers: SVM (0.7253), LR (0.6958), and XGBoost (0.7294).

C. Implications for Strategy Selection

The ordering B+C > A+C > A+B = A+B+C gives a direct rule for moderate binary imbalance on tabular data: with a strong ensemble, use class weighting on the original data first; use resampling alone as a fallback; and do not stack both unless the weights are recomputed on the post-resampling distribution. This mismatch is not specific to this dataset, because it follows from applying weights calibrated for one ratio to data balanced to another. Prior cost-sensitive work [5] points the same way, but here the claim is backed by a full four-way ablation rather than a single setting.

D. Limitations and Future Directions

The study uses one binary benchmark at moderate imbalance, so results under severe imbalance (minority ratio ≤ 10%) or multi-class settings remain open. XGBoost hyperparameters were fixed across tests, and recalibrating class weights on the post-resampling distribution was not tested; both are clear next steps.

VII. CONCLUSION

We ran a controlled ablation study on three strategies for reducing minority-class misclassification in imbalanced binary data: SMOTETomek resampling (A), class weighting (B), and XGBoost (C). Testing all four non-trivial combinations across five classifiers and six metrics allowed us to measure how much each strategy contributes on its own and what happens when they are stacked together.

The main finding is that class weighting on the original imbalanced data with no resampling applied, gave the best XGBoost results: macro-F1 of 0.7294 and ROC-AUC of 0.8306. The full three-strategy combination (A+B+C) matched Tests 1 and 4 exactly, confirming that adding the third strategy contributes nothing when the first two are already causing over-correction. What drives this pattern is a precision-recall tradeoff that goes the wrong way: when resampling and loss-level weighting both push toward the minority class at the same time, precision drops without any corresponding gain in recall, and macro-F1 ends up lower than with just one correction applied.

From these results, three practical takeaways emerge. First, stacking correction strategies does not guarantee better minority-class performance and can make it worse. Second, for tabular data at moderate imbalance, applying class weights to the original distribution is enough, and it beats all combined configurations tested here. Third, if resampling is part of the pipeline, class weights should be recalculated on the post-resampling distribution rather than the original one. All code and generated figures are made available to support reproducibility.

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